LSCSH: Lattice-Based Secure Cryptosystem for Smart Healthcare in Smart Cities Environment

Rajat Chaudhary, Anish Jindal, Gagangeet Singh Aujla, Neeraj Kumar, Ashok Kumar Das, and Neetesh Saxena

The authors propose a novel lattice-based secure cryptosystem for smart healthcare in future smart cities. LSCSH uses a lightweight key exchange and an authentication mechanism at several stages. The proposed scheme has been evaluated with respect to the communication and computation costs in comparison to the other competitive schemes. The results obtained prove the efficacy of the proposed scheme in comparison to the existing schemes.

Abstract

With the evolution of the Internet of Things, mutual sharing of information becomes easy among various smart devices across the globe. In such an environment, smart healthcare also needs to be expedited with an aim to provide remote diagnosis to patients located across the globe. However, as the sensitive information about a patient travels using an open channel (i.e., the Internet), there are chances of misusing this information. Hence, there is a requirement for a strong crypto mechanism for securely processing smart healthcare data. Most of the solutions reported in the literature use heavy computation-based solutions for providing security to smart healthcare data, which may not be suitable in view of less resources for modern smart devices. In this article, we propose a novel lattice-based secure cryptosystem for smart healthcare (LSCSH) in future smart cities. LSCSH uses a lightweight key exchange and an authentication mechanism at several stages. The proposed scheme has been evaluated with respect to the communication and computation costs in comparison to the other competitive schemes. The results obtained prove the efficacy of the proposed scheme in comparison to the existing schemes.

Introduction

The evolution of the Internet has paved the way for various new technologies, such as cloud computing, big data, quantum computing, and the Internet of Things (IoT) for providing quality of service (QoS) to end users. In an IoT environment, millions of devices communicate with each other using the Internet to sense, share, and process data. These *things* in IoT cover a wide variety of devices, such as sensors, smartphones, personal digital assistants, and laptops, to exchange information using a wired or wireless communication channel. According to a report by Gartner [1], there were around 8.4 billion connected devices in 2017. This is leading to expenditure and services worth US$2 trillion in sectors such as transportation, power, retail, logistics, smart cities, and healthcare. Among all these sectors, smart healthcare has been most revolutionized by the emergence of IoT due to the development of wearable bio-medical sensors. IoT in healthcare has been used to provide remote health monitoring, to study the effect of drug usage, and to offer elderly care using smart healthcare.

In smart healthcare, the sensors are deployed on the patients' side to continuously sense different parameters, such as heart rate, blood pressure, glucose level, and temperature, which are related to their health. The data gathered from various sensors is then transmitted to the healthcare provider, where it is stored in a database repository. This repository is then accessed by the doctor to monitor patients' conditions for providing treatment even from remote locations. IoT-based smart healthcare decreases the medical costs and increases the quality of life of a human being [2]. The healthcare data of patients is heterogeneous in nature and contains huge volumes of information, which can be in the form of signals, text, speech, and images. Thus, storing such data in a traditional database is quite a challenging task because of its high operational cost. Moreover, it could only be accessed locally by the doctors available in a hospital. Thus, cloud-based servers are often used to store and process patients' health records as they are cost-effective, scalable, and easier to manage [3].

However, sharing patients' sensitive information using IoT over the Internet leads to serious security and privacy concerns, as the smart healthcare system is vulnerable to cyber-attacks [3]. The exposure of such information to any third party can lead to data misuse or may create social discrimination against the patient on the basis of a medical ailment. Therefore, data access by an unauthorized person must be restricted by enforcing strong confidentiality mechanisms. Moreover, the data should be encrypted before transmitting as it would be stored on third-party cloud servers, which are also prone to cyber-attacks. Thus, in order to provide secure data transmission and storage in the smart healthcare environment, cryptographic mechanisms must be used to preserve privacy and evade cyber-attacks.

Motivation

Over the years, data privacy and security in healthcare applications have attracted a lot of attention from academicians and industry. There is sufficient literature available on the traditional security schemes used in smart healthcare applications. However, these mechanisms are not suitable for their direct applications while providing security in smart healthcare. For instance, the authors in [4] used an attribute-based encryption (ABE) scheme for storing the health records of

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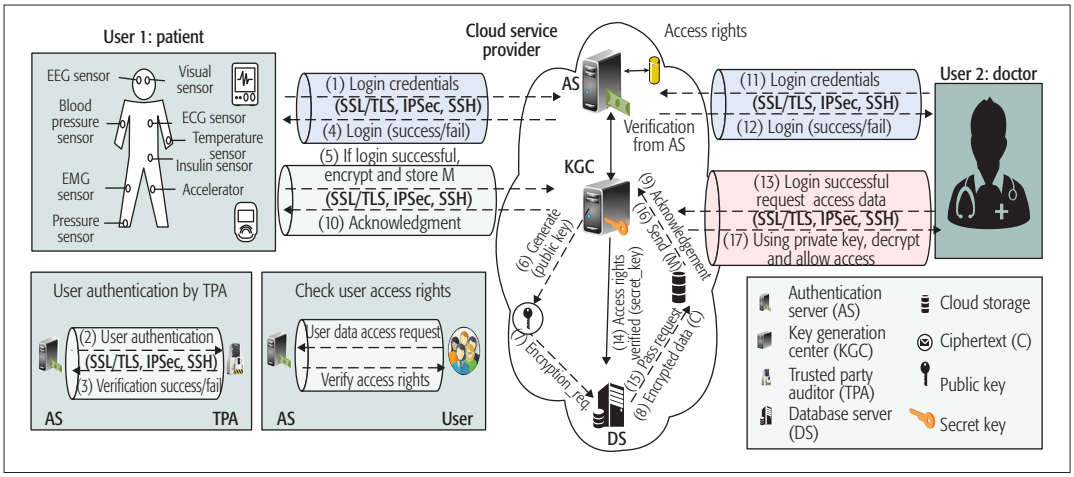

Figure 1. Steps involved in the proposed scheme.

patients on semi-trusted cloud servers. However, the general disadvantage of using ABE schemes is that these use bilinear operations having a high computation cost [5]. Apart from ABE, researchers have extensively used public key cryptographic systems, bilinear Diffie-Hellman (BDH), elliptic curve cryptography (ECC), ElGamal (EG), homomorphic encryption (HE), block cipher (BC) techniques, searchable encryption, or a combination of these techniques [2, 6–8]. However, there are many concerns related to these techniques. For example, public key cryptography such as Rivest, Shamir, & Adleman (RSA) depends on a large integer factorization on large abelian groups and therefore has slow processing speed. The ECC's security is based on a discrete logarithm and uses finite fields with smaller groups; however, its operations have a high computation cost. The EG algorithm is based on a cyclic group, and its security solely depends on the discrete logarithms in large prime domains.

Moreover, standard security methods including ABE, ECC, EG, and so on will become redundant in the near future as we are living in an era of quantum computers. These computers can easily solve such algorithms in polynomial time for detecting various security breaches. Thus, a security mechanism strong enough to withhold quantum attacks is required that can improve the future of smart healthcare applications. In this regard, lattice-based cryptography is a popular technique that is more resilient against quantum attacks while providing security in various application domains in the quantum age. This is because it cannot be deciphered in polynomial time even by a quantum computer and has been proven resilient against various known traditional cyber attacks [9]. The major threats that are more critical in smart healthcare applications, such as replay attack, perfect forward secrecy, chosen ciphertext attack, and man-in-the-middle attack, can also be handled effectively by using lattice-based cryptography.

Contributions

The major contributions of the proposed scheme are summarized as follows:

- A lattice-based mutual authentication scheme is designed that validates the requests between cloud storage and various end users such as patients and doctors.
- A data encryption scheme is formulated for data exchange between various users and cloud storage. This scheme is based on a third-party key exchange mechanism between users and the cloud to encrypt or decrypt the data.
- An access rights verification mechanism for granting permissions to end users is designed. This mechanism restricts the access rights of patients to their own data, while doctors can only view the information on patients who are linked to them.

Organization

The rest of the article is organized as follows. We present the system model of the proposed scheme. We describe the encryption, authentication, and access rights verification mechanisms in detail. The performance of the proposed scheme is evaluated and compared with existing schemes. Various open issues are discussed, and the work is concluded.

System Model

Figure 1 shows the system model used in the proposed scheme for communication between users (patients and doctors) and cloud service providers (CSPs). In this figure, there are three entities: user (user 1: patient and user 2: doctor), CSP, and trusted-party auditor (TPA). The patient's role in the proposed scheme is to send the data of body sensors or medical equipment to the CSP. Moreover, a patient can also view his/her historical data at a later stage. The doctor(s) can access the patients' data to monitor his/her health or suggest a line of treatment. The third entity, the CSP, consists of three servers: the authentication server (AS), key generation center (KGC), and database server (DS). The AS is responsible for authenticating the users and verifying their access rights with the help of TPA. The KGC helps to generate the public and secret keys. Finally, the DS performs the data encryption and decryption process based on the keys generated by the KGC. The authentication and encryption/decryption process are based on lattice-based cryptography, while the access rights are implemented using a tag matching scheme, discussed in the next section. The access rights are issued to patients and doctors such that a

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The public cloud is an insecure entity and it is risky to store the data in it in the plaintext form as high chances of active eavesdropping and man-in-the middle attacks can be launched by an adversary on public clouds. Hence, in order to secure the healthcare data, it is important to store the data in an encrypted form at the public cloud.

patient can only view his/her own data, while the doctor can view the details of only those patients' information who are being treated by him. The storage unit at the CSP is the public cloud unit where the encrypted data of the patients gets stored.

In the proposed scheme, the patient data is sensed by the IoT sensors or medical equipment, which gets stored in the public cloud for local and remote data access. The public cloud is an insecure entity, and it is risky to store the data in it in plaintext form as high chances of active eavesdropping and man-in-the middle attacks can be launched by an adversary on public clouds. Hence, in order to secure the healthcare data, it is important to store the data in encrypted form in the public cloud. To ensure secure communication between different parties involved in the communication, the secure channel is used by implementing traditional security protocols and tunneling techniques. These include secure socket layer (SSL), transport layer security (TLS), IP security (IPSec), and secure shell (SSH). The detailed steps involved during data transmission in the proposed scheme are shown in Fig. 1 and explained as follows.

Step 1: Initially, when the patient wants to store his/her data in the cloud, the login phase is initiated. The patient enters his/her credentials (i.e., an ID and password), and the login request is sent to the AS at the CSP.

Step 2: The AS verifies the login request with the help of TPA using an authentication mechanism as discussed.

Step 3: After the authentication process is completed, the TPA sends the notification to the AS regarding whether the login request is matched successfully or failed.

Step 4: The AS sends an acknowledgment to both the KGC and patient if the login is successful; otherwise, the AS sends only the login failure message to the patient.

Step 5: If the login is successful, the patient's request is passed further to the KGC for data encryption.

Step 6: The KGC generates a public key for the patient based on his/her credentials.

Step 7: The KGC sends the public key and the plaintext to the DS for data encryption.

Step 8: The DS takes input as plaintext and generates ciphertext as output by using the public key.

Step 9: The ciphertext gets successfully stored at the DS, and the notification is sent to the KGC.

Step 10: The KGC forwards the notification as an acknowledgment of the encrypted data to the user.

Steps 11, 12, 13: If the doctor (or patient itself) wants to access the data, the login credentials of the doctor (or patient) are verified by the TPA over a secure medium.

Step 14: Based on the access rights of the doctor (or patient), the KGC generates the secret key pair for the user and sends them to the DS as a request to access the data. The access rights are verified using a tag matching mechanism.

Step 15: The DS now decrypts the ciphertext using the received secret key pair.

Steps 16, 17: The plaintext produced is sent to the KGC, which sends the data to the user through the secure channel.

In this way, the data of patients is stored and accessed from the public cloud storage. The detailed description of the authentication, data encryption/decryption, and access rights phases are explained in the next section.

Lattice-Based Cryptosystem

Unlike the traditional classical computers where bits are represented in either 0 or 1 state, the design of the quantum computer is such that it uses bits in quantum state (known as qubits). These qubits are the combination of traditional 0 and 1 bits that work simultaneously. Moreover, only one spin direction, either vertical or horizontal, is chosen to identify the bits. The concept of qubits is more suitable for smart healthcare applications where it is necessary to deliver timely information to the end user. Apart from speed, the requirement of tight security is a major concern in smart healthcare applications, which can be ensured with the help of lattice operations. This post-quantum technique is suitable for public key cryptography security, which is based on the set of lattice operations. These operations are resilient against classical and quantum computer attacks [9]. They have already been used in the past by researchers to provide security in various application domains such as smart grid and authentication of RFID tags [10, 11]. Motivated by these facts, a lattice-based cryptosystem (LBC) is designed in the smart healthcare environment, as shown in Fig. 2.

Basics of Lattices

A lattice is denoted over the quotient ring of vector modulo, which performs faster matrix-vector arithmetic operations and fast Fourier transform for matrix-vector product represented in high-dimensions. In addition, an ideal lattice has no negacyclic codes over finite chain rings, which is based on block matrix structure to reduce the key storage space. Lattice vectors are resistant to quantum attacks, and breaking their security is equivalent to solving NP-hard problems (as more complex mathematical properties are used in lattices compared to RSA and ECC). Mathematically, the term *lattice* is defined as a regular ordered arrangement of isolated points in a two-dimensional plane. More precisely, a lattice $L(x)$ is a set of p linearly independent vectors $\{x^1, x^2, .., x^p\}$ defined in a ring R_p, where p is a prime number depicting the maximum degree of the polynomial vector [9]. The reason for defining a lattice in a ring R_p is that lattices are infinite in number, but lattice-based cryptography works on a finite space. The ring is a superset of group (Z) and abelian group (Z_p) with the additional properties of polynomial addition and multiplication [12]. The lattices are broadly classified into two types.

1. *Bound lattices*: In these lattices, the maximum value of an element is 1, and the minimum element value is 0 (i.e., $0 \le x \le 1$ defined in the form of matrix),
2. *Integer lattices*: It is a subgroup of a finite abelian group $(Z_p) \| Z_p/L$ whose vectors have integer coordinates. The essential parallelepiped structure of the lattice $L(x) \in R_p$ is defined as

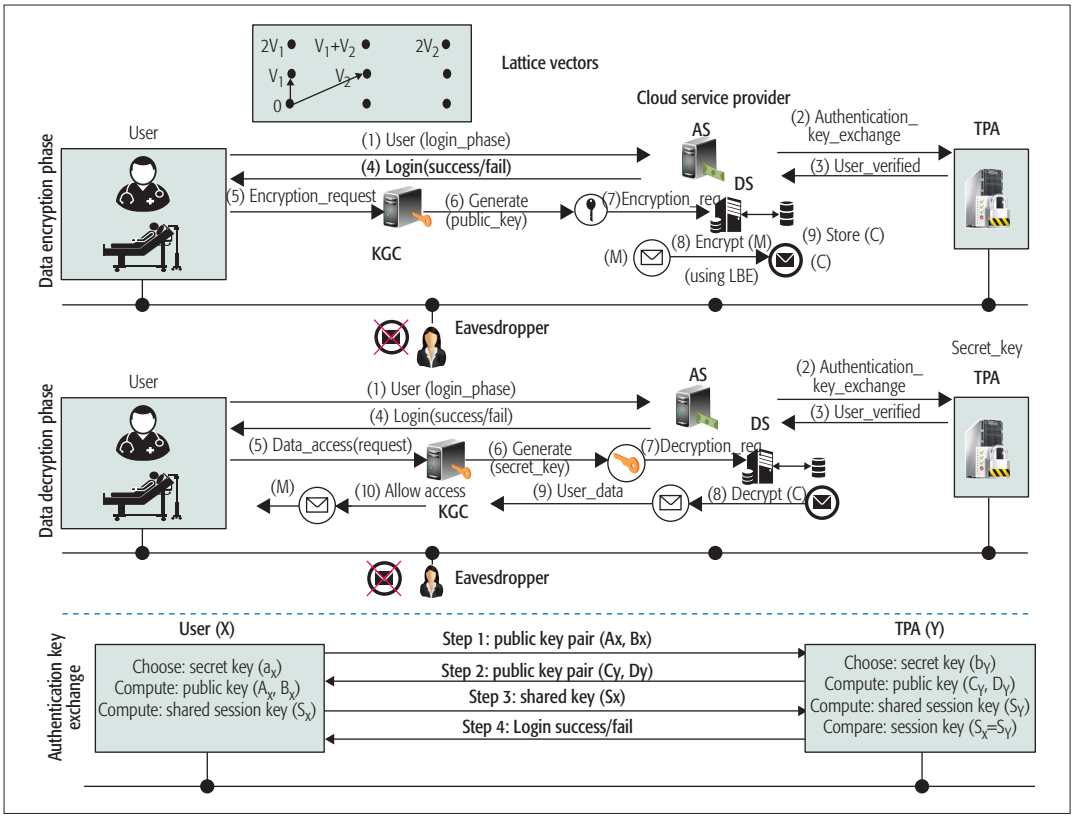


Figure 2. Lattice-based cryptosystem used in the proposed scheme.

Unlike the traditional classical computers where bits are represented in either 0 or 1 state, the design of the quantum computer is such that it uses bits in quantum state (known as qubits). These qubits are the combination of traditional 0 and 1 bits that work simultaneously. Moreover, only one spin direction either vertical or horizontal is chosen to identify the bits.

$$L(x) = L\left\{a_1.x^1, a_2.x^2, \ldots, a_i.x^i\right\}$$

$$L(x^1, x^2, \ldots, x^p) = \left\{\sum_{i=0}^{p-1}(a_i \times x^i) \middle| a_i \in [0,1] \in Z\right\} \quad (1)$$

where a_i is the coefficient of lattice $L(x)$, which belongs to the group Z.

An example of the encryption process using lattice-based polynomial vector is presented as follows. Suppose message m is to be encrypted and is represented in the polynomial form as $m(x) = 1 + x - x^2 + x^3 + x^4 - x^5$. The value of p is chosen to be 7, and the coefficient of x^6 in the vector $m(x)$ is 0. The matrix notation of the message $m(x)$ is represented as

$$m(x) = \begin{bmatrix} a_{11} & . & . & . & a_{1p} \\ . & . & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ a_{p1} & . & . & . & a_{pp} \end{bmatrix} \begin{bmatrix} x_{11} & . & . & . & x_{1p} \\ . & . & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ x_{p1} & . & . & . & x_{pp} \end{bmatrix}$$

$$c(x) = \left[\begin{array}{cccc|cccc} 0 & . & . & 0 & c_0 & c_1 & . & c_{p-1} \\ . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . \\ 0 & . & . & 0 & c_2 & c_1 & . & c_0 \\ 1 & . & . & 0 & s & 0 & . & 0 \\ 0 & 1 & . & 0 & 0 & s & . & 0 \\ . & . & . & . & . & . & . & . \\ 0 & 0 & . & 1 & s & . & . & s \end{array}\right] = \begin{bmatrix} 0 & c_i \\ I & sI \end{bmatrix} \quad (2)$$

(3)

where I is the identity matrix and s is an integer used in the encryption process.

Lattice-Based Encryption Scheme

The LBE scheme depends on a ring-based public key cryptosystem. The complete process of encryption and decryption works in four different phases, which are elaborated in Fig. 3 and described as below.

Setup Phase: Initially, the KGC chooses security parameters based on lattice polynomial vectors as an input. These parameters are shared between the KGC and DS for session initiation. Let $p = 5$ be a prime number chosen to define the maximum degree of the polynomial vectors with the coefficients −1, 0, and 1. The set of polynomials used in the encryption and decryption phases are $\{k(x), l(x), m(x), \text{and } d(x)\}$. These polynomial vectors are defined in such a way that the maximal degree of a polynomial should be 4, for example, $k(x) = -1 + X + 2X^3 - 7X^4$. When the degree of polynomial expression exceeds 4 in the multiplication operation, the polynomial equation is divided by $\{X^5 - 1\}$.

This reduces the multiplication degree of two polynomials to at most X^4, which in turn significantly reduces the overall computation cost. Based on the security parameters defined in the setup phase, the key generation phase is started.

Key Generation Phase: In this phase, the KGC generates the public and secret key pair, which is shared with the DS. For this purpose, the extended Euclidean algorithm is used to compute the value of $k_s(x)$. After this, the public key is calculated using the formula given in Fig. 3. The KGC also computes the secret key pair, which is sent to the DS at the time of decryption. The KGC transmits security parameters and the public key $e(x)$ to the DS using the secure channel with various security protocols (SSL, TLS).

The lattice-based authentication scheme is based on the enhancement of BDH key exchange protocol which suffers from forward secrecy attack while sharing the session key. To overcome the threat of such attacks, there is a need to derive a complex session key on both ends.

KEY GENERATION CENTER (KGC)	DATABASE SERVER (DS)
SETUP PHASE **Security parameters** Choose integers p, r, and s, polynomials set $\{k(x), l(x), m(x), d(x)\}$ with at most degree $(p\text{-}1)$, where p is a prime such that $p \geq 2$ r is co-prime to p, s is a power of 2 such that $gcd(p,s)=1$ and $gcd(r,s)=1$ with $s>r$, $k(x)$ is secret key vector, $l(x)$ is public key vector, polynomials vector. A lattice $L(x)$ is the combination of linearly independent vectors defined as below. $L(x) = \Sigma_{i=0}^{p-1} (a_i \times x^i \| a_i \in Z_p)$, where $\{a_i\}$ is the set of integers and $\{x^i\}$ is the set of vectors. Lattice is defined in convolution polynomial ring (R_p) in the form of $R = \frac{Z[x]}{(x^p-1)}, R_r = \frac{[Z/Zr][x]}{(x^p-1)}, R_s = \frac{[Z/Zs][x]}{(x^p-1)}$. where R is an $p \times p$ matrix having the basis vectors as columns The quotient ring of polynomial *modulo r* and *modulo s* and its vector coefficient are chosen in $\{-1,0,1\}$ i.e. $(-r/2$ to $+r/2)$ and $(-s/2$ to $+s/2)$.	
KEY GEN PHASE The secret key and the public key is computed as follows. Using the extended euclidean algorithm, $k(x)\, k_r(x) = 1(mod\ r)$ i.e. $k(x) = k_r^{-1}(x)(mod\ r)$, $k_r(x) = k^{-1}(x)(mod\ r)$, $k(x)\, k_s(x) = 1(mod\ s)$ i.e. $k(x) = k_s^{-1}(x)(mod\ s)$, $k_s(x) = k^{-1}(x)(mod\ s)$, where k_r and k_s are the inverses of k. **Public key:** $e(x) = [r\, k_s(x)\, l(x)](mod\ s)$ where $e(x)$ is an $p \times p$ matrix $\in R_p$. $\xrightarrow[\text{(secure channel) (SSL/TLS)}]{\langle security_parameters,\ e(x)\rangle}$	
	DATA ENCRYPTION PHASE **Input parameters**: $m(x)$ and $d(x)$ are polynomial vectors with maximal degree p-1, where $m(x) \in R_p$ and $d(x) \in R_p$. Compute ciphertext: $c(x) = [m(x) + d(x)\, e(x)](mod\ s)$ and store the encrypted data $c(x)$. $\xleftarrow[\text{(secure channel) (SSL/TLS)}]{\langle acknowledgement\rangle}$
Send the acknowledgement to the user When a new request is arrived for data access by the authentic user, compute a secret key. **Secret key** Choose the pair $(k(x), k_r(x))$, secret key = $\{k(x), kr(x)\}$ $\xrightarrow[\text{(secure channel) (SSL/TLS)}]{\langle secret\ key\rangle}$	
	DATA DECRYPTION PHASE **Input parameters**: $c(x)$ and $k(x)$, public parameters, $e(x)$. Compute plaintext: $t(x) = k(x)\, c(x)(mod\ s)$, $t(x) = k(x)\, [m(x) + d(x)\, e(x)](mod\ s)$, $t(x) = k(x)\, [m(x) + d(x)\, r\, k_s(x)\, l(x)](mod\ s)$, as $[k(x)\, k_s(x)] = 1(mod\ s)$. Finally, compute the plaintext using modulo r as $m(x) = [t(x)\ \ k_r(x)](mod\ r)$, $m(x) = [k(x)\, m(x)\, k_r(x)](mod\ r)$, and hence, it is equal to zero. Also, $gcd(k(x), k_r(x)) = 1(mod\ r)$, $m(x)(mod\ r) = m(x)$. $\xleftarrow[\text{(secure channel) (SSL/TLS)}]{\langle m(x)\rangle}$
KGS provides the original data $m(x)$ to the user in the plaintext mode.	Therefore, the required original plaintext polynomial with maximal degree p-1.

Figure 3. Data encryption and decryption phases based on lattice public-key cryptosystem.

Data Encryption Phase: After receiving the public key, the DS chooses the input parameters as message $m(x)$ and merges it with a random polynomial $d(x)$. As different sensors generate different types of data (or messages), $m(x)$ is first converted into binary format and represented in the matrix form of order $p \times p$. The ciphertext is then computed using the input parameters and the public key as described in Fig. 3. It is stored at the DS in the encrypted format, and the acknowledgment is sent to the KGC. When the user sends a new request to access the data, the KGC sends the secret key pair to the DS over a secure channel (SSL/TLS).

Data Decryption Phase: The DS computes the plaintext message using the input parameters $c(x)$, $k(x)$ and the secret key pair as shown in Fig. 3. Finally, the plaintext data is sent to the KGC, which forwards the data access request to the user over the secure channel.

Remark: LBCs like Nth degree truncated polynomial ring (NTRU) and learning with errors generate ciphertexts with greater lengths because of the polynomial multiplications with degree p. Hence, in order to reduce the additional burden on the network resources, the value of p is kept small in the proposed scheme. However, this value can be increased to create more complex ciphertext, thereby making it difficult for an adversary to break the ciphertext. Moreover, the message is stored in a two-dimensional matrix form that reduces the complexity of the overall encryption process. However, if the message is to be stored in higher dimension form or the degree of polynomial is large, the ciphertext can be shortened, and

USER (X)	TRUSTED PARTY AUDITOR (Y)
Public parameters G: multiplicative group and g: generator of G ($g \in G$). Choose a lattice $L(x)$ as a combination of p linearly independent vectors $\{a_0x^0, a_1x^1, ..., a_{p-1}x^{p-1}\} \in R_p$, where (a_i), (x^i) denotes the set of integers vectors. $R_p = \frac{Z_p[x]}{(x^p-1)}$. **Secret Key** (a_x) Let X choose a secret key vector $(a_x)\|(a_x) \in R_p$, where $a_x \leftarrow PW_x$ and PW_x is the password of X. **Public Key** (A_x, B_x) $A_x = (g^{(ax)})(mod\ p) \in R_p$. Let d_x be a random vector chosen by X. $B_x = g^{dx}\ (mod\ p) \in R_p$.	
$\xrightarrow[\text{(secure channel) (SSL/TLS)}]{\langle A_x, B_x, ID_x \rangle}$ where PW_y is the password of Y.	**Public parameters:** G, g, p, s, R_p. *Secret Key* (b_y) Suppose Y chooses a random vector $b_y\|b_y \in R_p$ as a secret key. $b_y \leftarrow PW_y$, **Public Key** (C_y, D_y) $C_y = g^{(by)}$ (mod p) Suppose d_y be a random vector chosen by Y. $D_y = (g^{(dY)})$ (mod p) $\in R_p$ $u = h_1(ID_x\|ID_y\|B_x)$ (mod p) $\in R_p$ $v = h_2(ID_x\|ID_y\|B_x\|D_y)$ (mod p) $\in R_p$, where (h_1, h_2) are the hash functions.
Session Key $u = h_1(ID_x\|ID_y\|B_x)$(mod p) $\in R_p$. $S_x = [(C_y)^v + D_y]^{(a_x \cdot u+d_x)}\ (mod\ p)$ $\xrightarrow[\text{(secure channel) (SSL/TLS)}]{\langle S_x \rangle}$	$\xleftarrow[\text{(secure channel) (SSL/TLS)}]{\langle C_y, D_y, ID_y, v \rangle}$ **Session Key** $S_y = [(A_x)^u + B_x]^{(b_y \cdot v+d_y)}$ (mod p) **Session Key Matching** $S_x = [(C_y)^v + D_y]^{(a_x \cdot u+d_x)}$ (mod p) $S_x = [g^{(b_y \cdot v)} + g^{(dY)}]^{(a_x \cdot u+d_x)}$ (mod p) $S_x = [(A_x)^u + B_x]^{(b_y \cdot v+d_y)}$ (mod p) $S_x = S_y$ (session key matches successfully).

Figure 4. Secure authentication key exchange dialogues based on lattice public-key cryptosystem.

complexity can be further reduced by using techniques like the dimension-modulus reduction scheme [12].

Lattice-Based Authentication Scheme

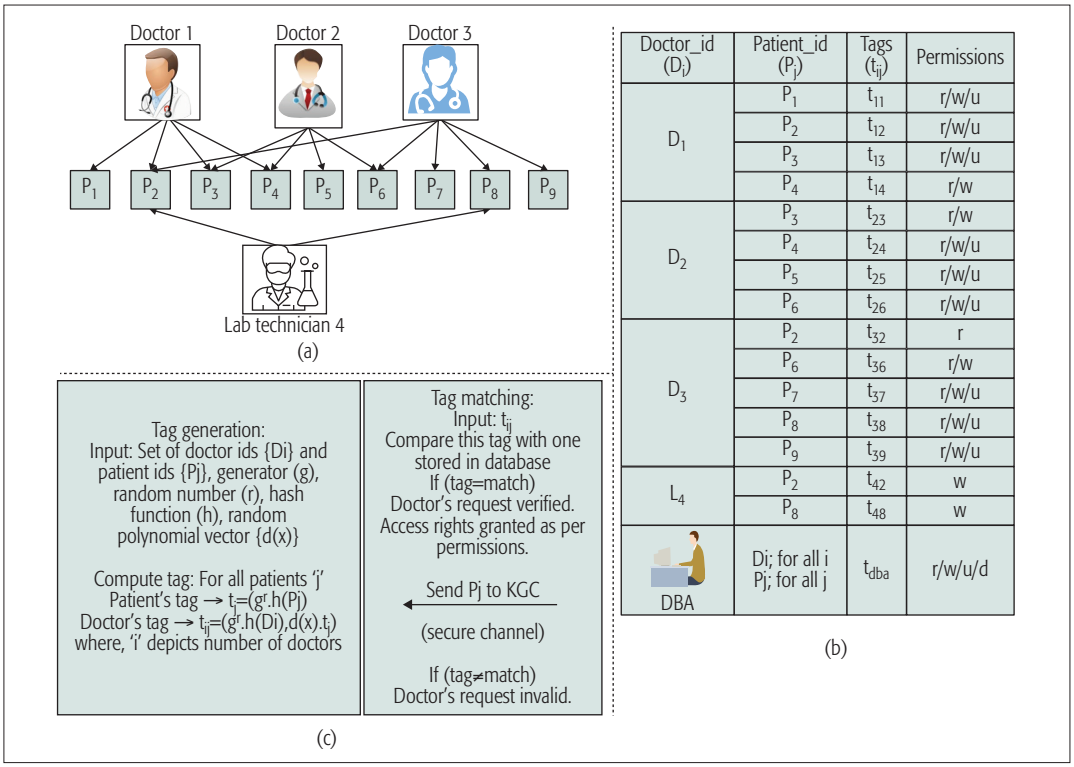
The validation of the credentials by the user is authenticated by using the lattice-based authentication scheme shown in Fig. 2. The lattice-based authentication scheme is based on the enhancement of the BDH key exchange protocol, which suffers from forward secrecy attack while sharing the session key. To overcome the threat of such attacks, there is a need to derive a complex session key on both ends. This key is computed using lattice-based vector equations (which are resilient against such attacks) in the proposed scheme. The complete procedure used in this authentication scheme is shown in Fig. 4. As shown in the figure, initially the user randomly chooses a secret key (a_X) on the basis of his/her password. Then this secret key is used to compute the public key pair $<A_X, B_X>$, which is sent to the TPA over a secure medium along with the user ID (ID_X). The TPA generates a public key pair of its own $<C_Y, D_Y>$ and two polynomials (u, v). The public key pair along with one of the calculated polynomials (i.e., v) is sent to the user. Using this key pair, the user computes a session key (S_X) using his/her identity information, which is sent to the TPA. The TPA also computes a session key (S_Y) whose value is compared with the session key received from the user. If the two values are matched, the user is deemed authentic. It is worth mentioning that the session keys are time-dependent values and expire after some time [13]. Thus, if an adversary tries to attack the system using these keys at a later stage, the attack would fail.

Access Right Verification Mechanism

The access right depicts the set of permissions for the users to access the information of the patients. The considered permissions are r/w/u/d, which depict read, write, update, and delete, respectively. The access rights are assigned by the hospital's database administrator (DBA) when the patient registers at the hospital. The patients are assigned only read permissions on their rights so that they can view only their own information, while doctors can have multiple permissions on their access rights to view information of multiple patients. An example scenario is considered in Fig. 5a to elaborate the concept of access rights. In this figure, multiple doctors (including a lab technician) associated with multiple patients are considered. As depicted in Fig. 5a, more than one doctor is associated with multiple patients (e.g., P_2). This type of scenario occurs when a patient has to consult more than one doctor. For example, in the case of a chronic disease, it might be possible that a patient is examined by a surgeon but also needs to consult with a neurologist as well as lab technician (to generate reports of tests) at the same time. In this case, doctor 1 has all the permissions

The access rights are assigned by the hospital's database administrator (DBA) when the patient registers at the hospital. The patient is assigned only read permissions on his rights so that they can view only their own information while a doctor can have multiple permissions on their access rights to view information of multiple patients.

The bandwidth and network requirements for the smart healthcare network can be optimized in order to increase its QoS. Moreover, it would also help to provide timely response to the queries of doctors and patients, which can be a life-saving factor in critical situations.



Doctor_id (D_i)	Patient_id (P_j)	Tags (t_{ij})	Permissions
D_1	P_1	t_{11}	r/w/u
	P_2	t_{12}	r/w/u
	P_3	t_{13}	r/w/u
	P_4	t_{14}	r/w
D_2	P_3	t_{23}	r/w
	P_4	t_{24}	r/w/u
	P_5	t_{25}	r/w/u
	P_6	t_{26}	r/w/u
D_3	P_2	t_{32}	r
	P_6	t_{36}	r/w
	P_7	t_{37}	r/w/u
	P_8	t_{38}	r/w/u
	P_9	t_{39}	r/w/u
L_4	P_2	t_{42}	w
	P_8	t_{48}	w
DBA	Di; for all i Pj; for all j	t_{dba}	r/w/u/d

Figure 5. Access rights verification policy.

(except delete), and doctor 3 has only read permission, while the lab technician has only write permission to upload the lab record as shown in Fig. 5b.

In addition, when the DBA assigns the patient(s) to the doctor(s), a tag value for each patient-doctor pair is calculated by the system. The steps used in the generation of this tag value are presented in Fig. 5c. This tag value is associated with the doctor to view the set of permissions for the particular patient that are assigned to the doctor. For the example scenario of Fig. 5a, the corresponding tag value and associated set of permissions are shown in Fig. 5b. This tag value is used by the doctor to view his/her access rights while accessing the data of patient(s). Now, when the doctor wants to access the data of a particular patient, its tag value is sent to the *AS*. The *AS* matches the received value with the value stored in its database. If the two values are matched, the doctor is granted access to the patient's data based on his/her permissions, and the patient's id is sent to the KGC for generating its secret key to decrypt its data. The steps involved in tag matching are given in Fig. 5c. It is to be noted that the access rights of patients and doctors can be added or revoked only by the DBA, who has the permissions to change any information, as depicted in Fig. 5b.

Performance Evaluation

In this section, we evaluate the performance of the proposed scheme with respect to communication and computation costs required during the secure authentication key exchange dialogues shown in Fig. 4. After that, we compare the functionality features of the proposed scheme with other relevant existing schemes.

Computation Cost

For computing the computation cost of the proposed scheme during the authentication key exchange dialogues, let T_{exp} and T_h denote the time needed for executing a 1024-bit modular exponentiation and a one-way hash function, respectively. Based on the experimental results report in [14], we have $T_{exp} \approx 3.85$ ms and $T_h \approx 0.0023$ ms. The computation costs required for user X and trusted party auditor Y are $4\ T_{exp} \approx 15.4$ ms and $11\ T_{exp} + 2\ T_h \approx 42.3546$ ms, respectively. Thus, the total computation cost becomes $15\ T_{exp} + 2\ T_h \approx 57.7546$ ms.

Communication Cost

For analyzing the communication cost of the proposed scheme during the authentication key exchange dialogues, we assume that the message size is 1024-bits so that the modular exponentiation needs 1024 bits in order to provide sufficient security of the scheme. The message digest (hash output) is 160 bits if we assume the Secure Hash Standard (SHA-1) algorithm [15], and the identity is 128 bits. The messages $\langle A_X, B_X, ID_X \rangle$, $\langle C_Y, D_Y, ID_Y, v \rangle$, and $\langle S_X \rangle$ require (1024 + 1024 + 128) = 2176 bits, (1024 + 1024 + 128 + 1024)= 3200 bits and 1024 bits. The communication costs required for user X and trusted party auditor Y are 3200 bits each. Hence, the total communication cost turns out to be 6400 bits.

Comparative Analysis

The comparison of the proposed lattice-based scheme with a few of existing schemes in the literature available for smart healthcare application is carried out in terms of its functionality. The brief summary of this comparison is given in Table 1. It can be inferred from this table that the scheme in [2] does not consider F2, F3, F6, and F7, and has

not explored F8 and F9. The scheme of Li *et al.* [4] did not explore F4, F8, F9, and F10, and lacks security in F6. The scheme of Yang *et al.* [6] is lacking in F2, F5, and F6, and has not explored F8, F9, and F10. Similarly, Ara *et al.* [7] have not considered F2, F5, F6, and F9; while Huang *et al.* [8] have not accounted for F2, F4, F6, and F8. However, the proposed scheme is resilient against all kinds of attacks and is able to provide maximum security in the smart healthcare network.

Open Issues and Future Directions

In the smart healthcare network, security is one of the major concerns and has been addressed in the present work. However, there are still various open issues that need new and efficient solutions in the future to further improve the performance of smart healthcare applications. A few such issues are listed below:

Communication network: The bandwidth and network requirements for the smart healthcare network can be optimized in order to increase its QoS. Moreover, it would also help to provide timely response to the queries of doctors and patients, which can be a life-saving factor in critical situations.

Complexity trade-off: The higher order polynomial operations in LBC need more computation time. Thus, one needs to decide the order of polynomial vectors so that it takes less computation time without jeopardizing the security of the system.

Data Duplication: It is often observed that sensing the data in close time intervals leads to similar values of data. Thus, storing same data values would unnecessarily increase the size of data storage and corresponding security mechanisms. Hence, storing duplicate values of data should be avoided.

Fault tolerance: As in most smart healthcare applications, wireless body sensor networks are used. The reliability and fault tolerance of sensor nodes are other aspects that need to be given more emphasis.

Conclusion

With an evolution of Internet-enabled smart devices, smart healthcare has become reality in which end users receive medical treatment from anywhere across the globe. However, as the information among the different entities such as end users, doctors, and hospitals is shared using an open channel (i.e., the Internet), there are higher chances of information leakage, which may affect the implemented solution in this environment. Moreover, it is a challenging task to design an effective cryptographic solution for such an environment keeping in view the resource constrained nature of the smart devices used in this environment. Hence, to cope with these issues, in this article, a lattice-based secure cryptosystem for smart healthcare in the smart cities environment is presented. The proposed scheme incurs low computation and communication costs as compared to the existing solutions.

In the future, we will explore more about the channel security and cryptographic primitives for healthcare system domain data in a distributed environment.

Moreover, dimension reduction techniques can also be incorporated in the proposed scheme to reduce the ciphertext size of higher lengths as well as the message decryption complexity.

Features	Gope-Hwang [2]	Li *et al.* [4]	Yang *et al.* [6]	Ara *et al.* [7]	Huang *et al.* [8]	Proposed
F0	BC	ABE	BDH	EG	HE	LBE
F1	✓	✓	✓	✓	✓	✓
F2	×	✓	×	×	×	✓
F3	×	✓	✓	✓	✓	✓
F4	✓	–	✓	✓	–	✓
F5	✓	✓	×	×	✓	✓
F6	×	×	×	×	×	✓
F7	×	✓	✓	✓	✓	✓
F8	–	–	–	✓	–	✓
F9	–	–	–	–	✓	✓
F10	✓	–	–	✓	✓	✓
F11	✓	✓	✓	✓	✓	✓
F12	✓	✓	✓	✓	✓	✓

F0: cryptographic technique used; F1: mutual authentication; F2: access right verification; F3: strong data encryption; F4: man-in-the-middle attack; F5: session key management; F6: resilient against quantum computing; F7: data stored in encrypted form; F8: known ciphertext attack; F9: known plaintext attack; F10: data masquerading attack; F11: replay attacks; F12: privacy preservation.

✓: Represents that the particular scheme is secure; ×: Represents that particular scheme is insecure; –: Authors did not elaborate on the aspect in their work.

Table 1. Comparison with existing schemes.

References

[1] Gartner, Inc., "Gartner Says 8.4 Billion Connected 'Things' Will Be in Use in 2017, Up 31 Percent From 2016"; http://www.gartner.com/newsroom/id/3598917, accessed June 2017.

[2] P. Gope and T. Hwang, "BSN-Care: A Secure IoT-Based Modern Healthcare System Using Body Sensor Network," *IEEE Sensors J.*, vol. 16, no. 5, 2016, pp. 1368–76.

[3] J. Zhou *et al.*, "Security and Privacy for Cloud-Based IoT: Challenges," *IEEE Commun. Mag.*, vol. 55, no. 1, Jan. 2017, pp. 26–33.

[4] M. Li *et al.*, "Scalable and Secure Sharing of Personal Health Records in Cloud Computing Using Attribute-Based Encryption," *IEEE Trans. Parallel and Distrib. Systems*, vol. 24, no. 1, 2013, pp. 131–43.

[5] S. Patranabis, Y. Shrivastava, and D. Mukhopadhyay, "Provably Secure Key-Aggregate Cryptosystems with Broadcast Aggregate Keys for Online Data Sharing on the Cloud," *IEEE Trans. Computers*, vol. 66, no. 5, 2017, pp. 891–904.

[6] Y. Yang, X. Zheng, and C. Tang, "Lightweight Distributed Secure Data Management System for Health Internet of Things," *J. Network and Computer Applications*, vol. 89, 2017, pp. 26–37.

[7] A. Ara *et al.*, "A Secure Privacy-Preserving Data Aggregation Scheme Based on Bilinear ElGamal Cryptosystem for Remote Health Monitoring Systems," *IEEE Access*, vol. 5, 2017, pp. 12,601–17.

[8] H. Huang *et al.*, "Private and Secured Medical Data Transmission and Analysis for Wireless Sensing Healthcare System," *IEEE Trans. Industrial Informatics*, vol. 13, no. 3, 2017, pp. 1227–37.

[9] O. Regev, "On Lattices, Learning with Errors, Random Linear Codes, and Cryptography," *JACM*, vol. 56, no. 6, article 34, 2009.

[10] A. R. Abdallah and X. S. Shen, "Lightweight Lattice-Based Homomorphic Privacy-Preserving Aggregation Scheme for Home Area Networks," *6th Int'l. Conf. Wireless Commun. and Signal Processing*, Hefei, China, 2014, pp. 1–6.

[11] E. El Moustaine and M. Laurent, "A Lattice Based Authentication for Low-Cost RFID," *IEEE Int'l. Conf. RFID-Technologies and Applications*, Nice, France, 2012, pp. 68–73.
[12] Z. Brakerski and V. Vaikuntanathan, "Efficient Fully Homomorphic Encryption from (Standard) LWE," *SIAM J. Computing*, vol. 43, no. 2, 2014, pp. 831–71.
[13] J. Ding *et al.*, "Provably Secure Password Authenticated Key Exchange Based on RLWE for the Post-Quantum World," *Cryptographers' Track at the RSA Conf.*, 2017, pp. 183–204.
[14] V. Odelu, A. K. Das, and A. Goswami, "An Efficient Biometric-Based Privacy-Preserving Three-Party Authentication with Key Agreement Protocol Using Smart Cards," *Security Commun. Networks*, vol. 8, no. 18, 2015, pp. 4136–56.
[15] NIST, Secure Hash Standard, FIPS PUB 180-1, U.S. Dept. of Commerce, Apr. 1995.

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